

THE EXTREME CONDITIONS CATALYTIC CELL FOR BL01 AT ALBA

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Abstract

A new catalytic cell has been developed for the Infrared Spectroscopy and Microscopy (MIRAS-BL01) beamline at the ALBA synchrotron. The aim of this instrument is to study catalytic reactions, crucial for advancing sustainable chemistry by enabling energy-efficient processes and minimizing by-products. Infrared (IR) spectroscopy offers key molecular insights, helping identify active species, understand mechanisms and link structure to activity. It also monitors catalysts in real time, revealing structural changes that affect performance.

The reactor is designed to operate in transmission mode from vacuum conditions to pressures up to 20 bar of different mixtures of gases and within a wide temperature range, covering from cryogenic temperatures up to a maximum of 500°C, while allowing the sample to move vertically few millimetres in order to alternate between exposing it and the background.

Currently in production, the design's key aspects are presented, covering the sample position mechanics, the various FEA calculations performed as well as the necessary auxiliary systems, such as cooling mechanisms and the pressurized gas circuit.

INTRODUCTION

The MIRAS-BL01 is a beamline devoted to Fourier Transform Infrared (FTIR) spectroscopy and microscopy. It consists on a modern synchrotron-based infrared spectrometer and microscope capability covering a wavelength range optimised for investigations from Mid-IR (700-4000 cm⁻¹) to the Far-IR (80-600 cm⁻¹) spectral ranges.

One of the potential research fields in this beamline is the study of catalytic reactions. A preliminary experiment carried out in collaboration with the Instituto de Tecnología Química (ITQ) in Valencia employed a bolometer detector in the far-infrared range and tested different gases under multiple configurations and modes. This cell provided valuable insights that inspired the development of the present design. The following research stage will focus on the development of a transmission-mode approach.

For this reason, a new catalytic cell has been designed for MIRAS aimed at advancing studies in this field.

TECHNICAL SPECIFICATIONS

The specifications that the cell must fulfil are listed below:

- To hold a sample pellet of diameter Ø=13 mm and thickness t=0,025 mm.
- Vertical movement of the sample around 20 mm to be exposed or not to the beam.
- Sample temperature from -196 °C to 500 °C.
- Possibility to work under vacuum (10⁻⁶ mbar) or accommodate a gas mixture (O₂, H₂, CO, and an inert gas like Ar) under 20 bar maximum working pressure and under flow mode.
- Easy integration inside the spectrometer (BRUKER Vertex 70).
- Volume of the setup minimized as much as possible.

SYSTEM DESCRIPTION

The proposed catalytic cell solution is shown in Fig 1. This cell consists on three subsystems. The first, called the heat exchanger, is responsible of the sample heating and cooling. The second is the sample stick, which includes the sample holder and the vertical motion mechanism. Finally, the pressure vessel contains the sample at the required pressure. Each of these components is described in detail in the following sections.

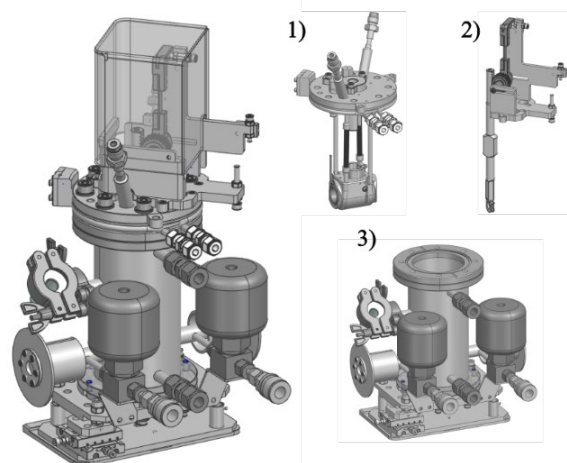


Figure 1: Left: general view of the catalytic cell. Right: 1) Heat exchanger. 2) Sample stick. 3) Pressure vessel.

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Heat Exchanger

This part consists on the components shown in Fig. 2. The main body is 3D-printed in AISI 316L stainless steel. The piece contains an opening through which the beam will pass across the sample. Around this opening, one of the heaters integrated into the cell is inserted. The Inconel 600 heater (Thermocoax) is able to provide a maximum power of 450 W. In addition, the piece contains an internal channel for circulating cold nitrogen gas or liquid nitrogen. It also includes a K-type thermocouple for temperature monitoring and control. The component is attached to a custom top flange equipped with the necessary electrical and fluid feedthroughs. Due to the high temperatures that can be reached in certain cases, this top flange is provided with a surrounding tube for water cooling.

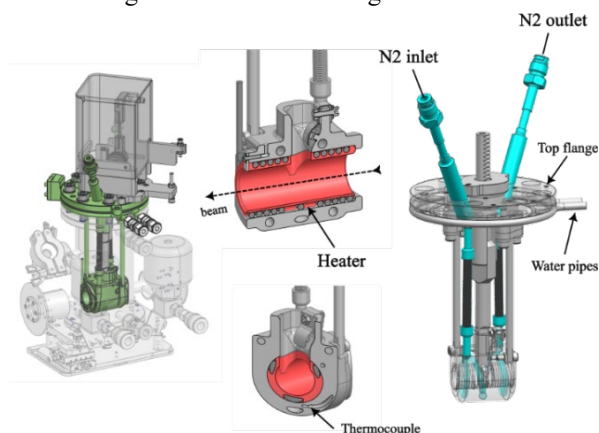


Figure 2: Heat exchanger system. Left: Highlighted system in the cell. Middle: Heater and thermocouple inside the heat exchanger body. Right: Detail of N₂ cooling circuit and water pipes.

Sample Stick

The sample stick operates as the sample holder of the catalytic cell. As shown in Fig. 3, it is inserted through the top flange until reaching the interior of the heat exchanger, where the sample is positioned in the beam path. Inside the stick, a second heater cartridge is installed, located as close as possible to the sample to ensure efficient heating. A second K-type thermocouple is positioned adjacent to the heater to provide precise temperature monitoring. This heater is capable of delivering a maximum power of 50 W.

One of the design requirements was to enable a vertical displacement of the sample by 20 mm, allowing the beam to pass without interacting with the sample while preserving the established pressure conditions. To achieve this requirement, the mechanism illustrated in Fig. 4 was implemented.

It consists of a gear system which, through manual actuation, enables 20 mm vertical displacement of the sample with a simple 90-degree rotation of the lever. The lever is made of a spring leaf which, when the sample is inserted, acts as a preload for the sample stick, ensuring improved contact between the base of the sample stick and the heat exchanger. Internally another spring pushes the sample tip to the heat exchanger face in order to improve as well the thermal contact.

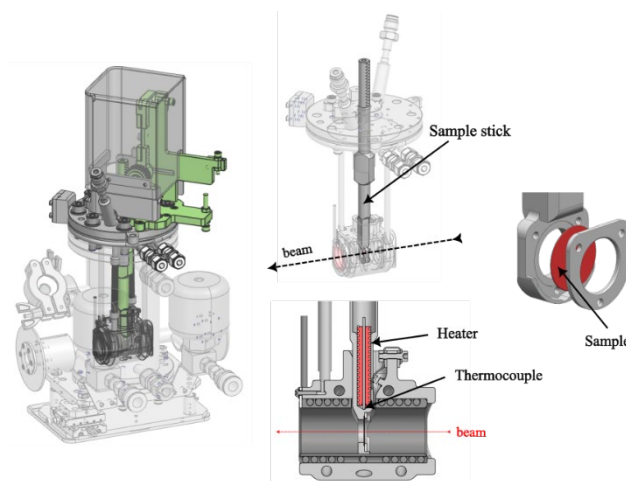


Figure 3: Sample stick system. Left: Highlighted system in the cell. Middle: Sample stick through top flange and inside heat exchanger body. Detail of heater and thermocouple. Right: Detail of the sample holder.

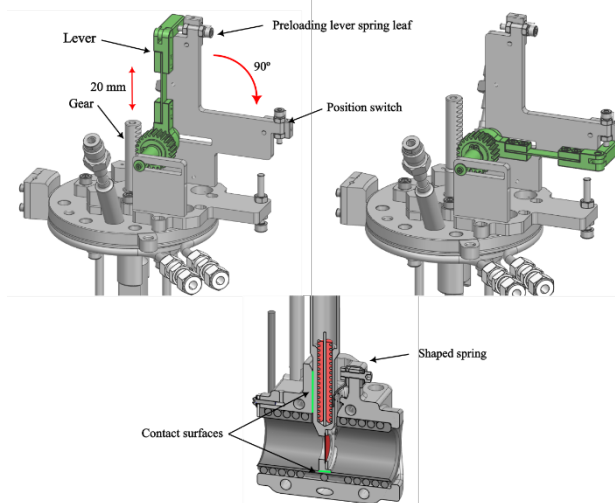


Figure 4: Sample vertical movement mechanism. Top-left: Sample down position. Top-right: Sample up position. Bottom: Preloaded thermal contact surfaces.

Pressure Vessel

The final element described is the pressure vessel. This vessel features a CF63 interface to allow the insertion of the entire system described above from the top. Upstream and downstream the beam axis, it is equipped with ports fitted with custom viewports for the beam to pass through. These viewports are made using an infrared transparent material as Si, ZnSe or CaF₂ windows, with thicknesses of 2 mm and 4 mm, which are interchangeable depending on the experimental requirements.

The vessel is additionally equipped with dedicated ports for the inlet and outlet of the process gas mixture, as well as a separate port for connection to a vacuum system. The central cylinder incorporates a double-wall construction designed to allow the circulation of cooling water, ensuring thermal management during experiments conducted under extreme temperature conditions. Fig. 5 shows a general

view of the vessel, together with more details on the custom viewports and the double-wall cooling system.

Figure 5 also illustrates the O-ring arrangement employed to maintain leak-tightness during the insertion and the manipulation of the sample stick.

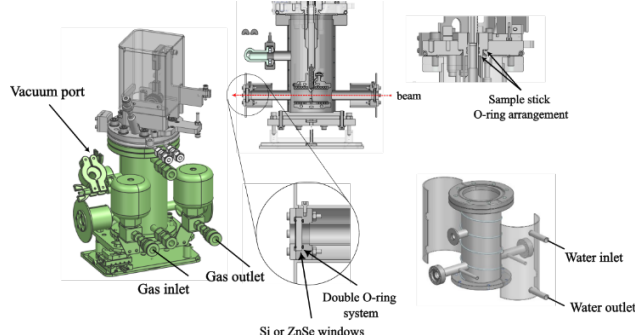


Figure 5: Pressure vessel. Left: Highlighted system in the cell. Details for gas and vacuum ports. Top-middle: View-section of the vessel, showing beam path. Bottom-middle: Detail on custom viewport. Top-right: Sample stick O-ring arrangement. Bottom-right: Double-wall cooling system.

CALCULATIONS

An extensive series of simulations were done for the catalytic cell to meet the specified requirements, ensure operational safety and account for the maximum power consumption of the heaters. In addition, numerous finite element analyses (FEA) were performed to assess the system's mechanical, thermal, and fluid-dynamic performance. Although this paper presents the most relevant thermal results, the complete and extensive study can be consulted in the referenced work [1].

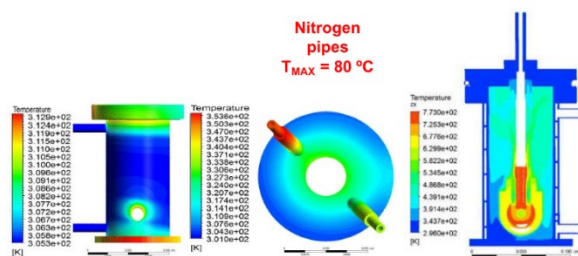


Figure 6: Thermal results for extreme conditions.

The calculation considered most relevant involves the most extreme operating conditions for the cell: a maximum sample temperature of 500 °C combined with a maximum gas pressure of 20 bar in the vessel (argon gas considered). Figure 6 shows some of the results. It can be observed that while the sample temperature is around 480 °C, the surrounding areas of the cell do not exceed 40 °C, except for the nitrogen pipes, where maximum temperature of 80 °C is reached. Additional protection will therefore be required for these components.

It should be noted that these results correspond to steady-state conditions. Empirical tests with the cell will be carried out to evaluate the transient behaviour of both the sample temperature and the temperature of the chamber surroundings, by varying parameters such as the heater power outputs and the cooling water flow rate.

ANCILLARY EQUIPMENT

In addition to the primary components of the catalytic cell, several auxiliary systems are required to ensure safe and reliable operation under the future experimental conditions. These include a mass flow controller for precise regulation of the process gas mixture, a gas panel engineered to operate at the required flow and pressure and equipped with safety devices to protect both the system and the operators, a power supply controller for the heaters to deliver and regulate the required thermal input, a liquid nitrogen dewar supplying cold nitrogen gas or liquid nitrogen for the cryogenic circuits and a water chiller providing controlled cooling to the pressure vessel and top flange, particularly during high-temperature or long-duration experiments. Together, these auxiliary elements are critical to maintaining the stability, safety, and performance of the catalytic cell during operation.

CONCLUSIONS

The here exposed work, which has presented several engineering challenges, introduces a solution aimed at fulfilling all specified requirements, with the expectation that it will contribute to advancing research in the field of infrared-based catalysis. Addressing these challenges required an extensive finite element analysis (FEA) study, involving more than 52 simulations, some of which comprised models exceeding 25 million elements. The system is currently in production and is expected to be operational at MIRAS beamline by fall 2025 or early 2026.

REFERENCES

- [1] A. Jacquet *et al.*, "Thermal simulations of a new experimental chamber for catalytic reaction studies at the MIRAS beamline of the ALBA synchrotron," *J. Phys. Conf. Ser.*, vol. 3010, no. 012111, pp. 1-6, May 2025.
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